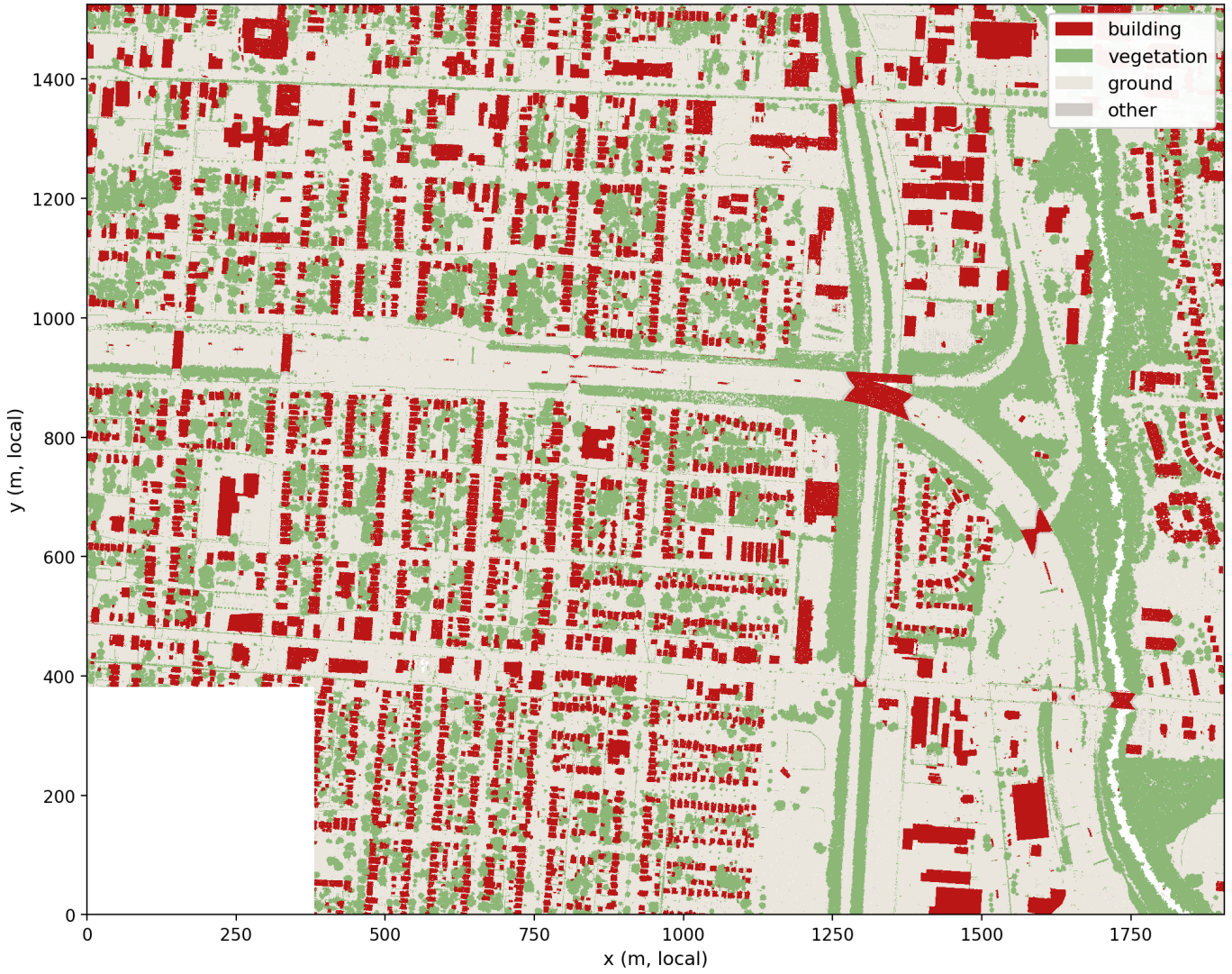
Last time: semantic segmentation with RandLA-Net



- 2.76 km² east Columbus, OH
- 19 USGS 2019 tiles · 76.35 M points (~30 pts/m²)
- 4 classes: ground / vegetation / building / other
- building IoU 0.978 (training tiles, checkpoint metric)
- → 8.6 M building points (11.3 %) = input to this stage

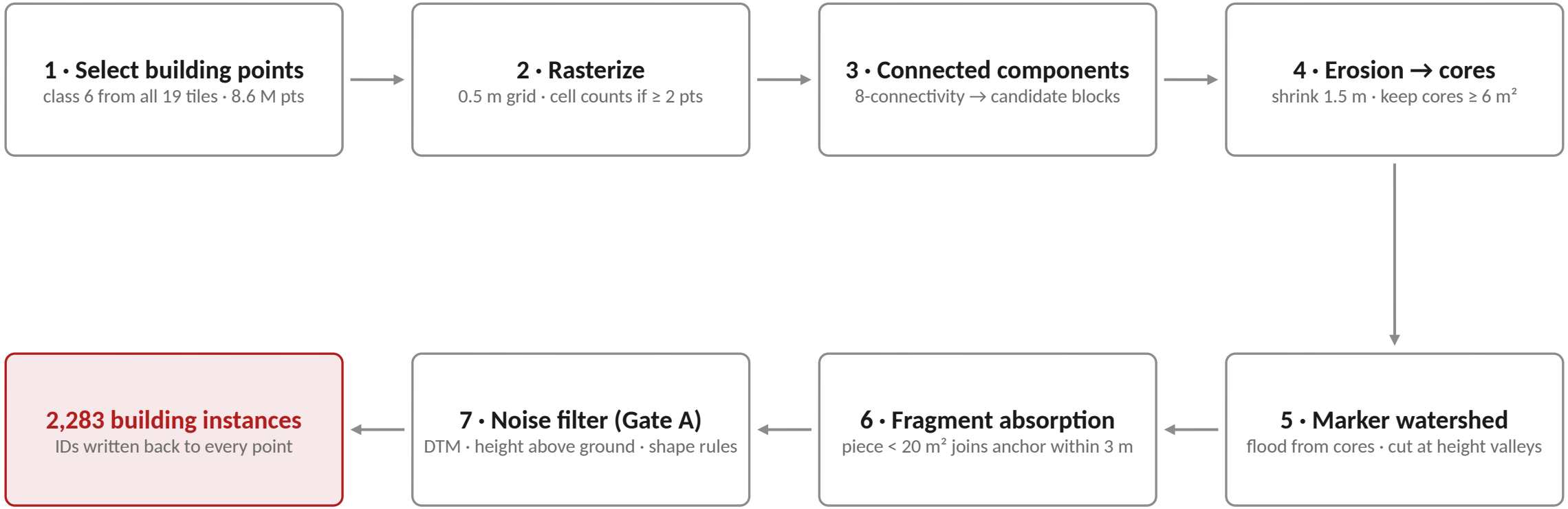
This stage: split the building points into individual buildings



Goal: one instance ID per building.

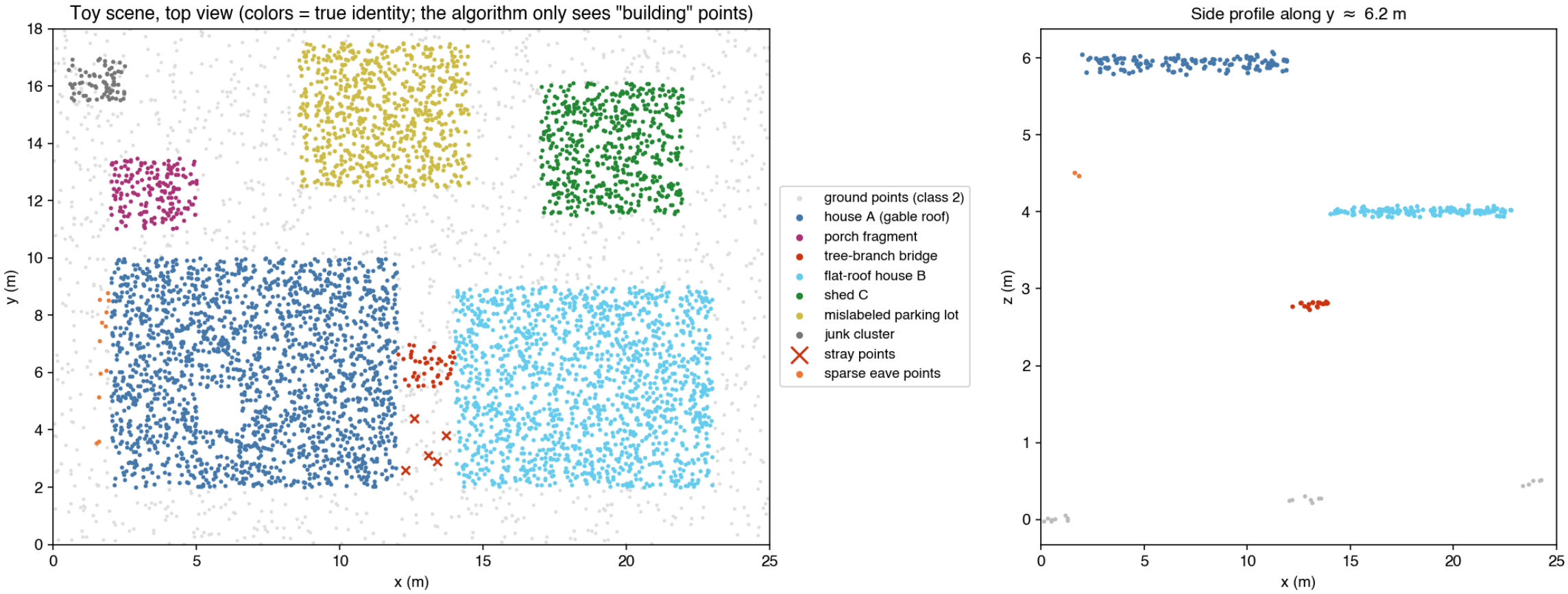
- tree bridges weld neighboring houses together
- occlusion fragments split one house apart
- parking lots, bridges, power lines hide in the mask

Method: a 2.5-D raster pipeline



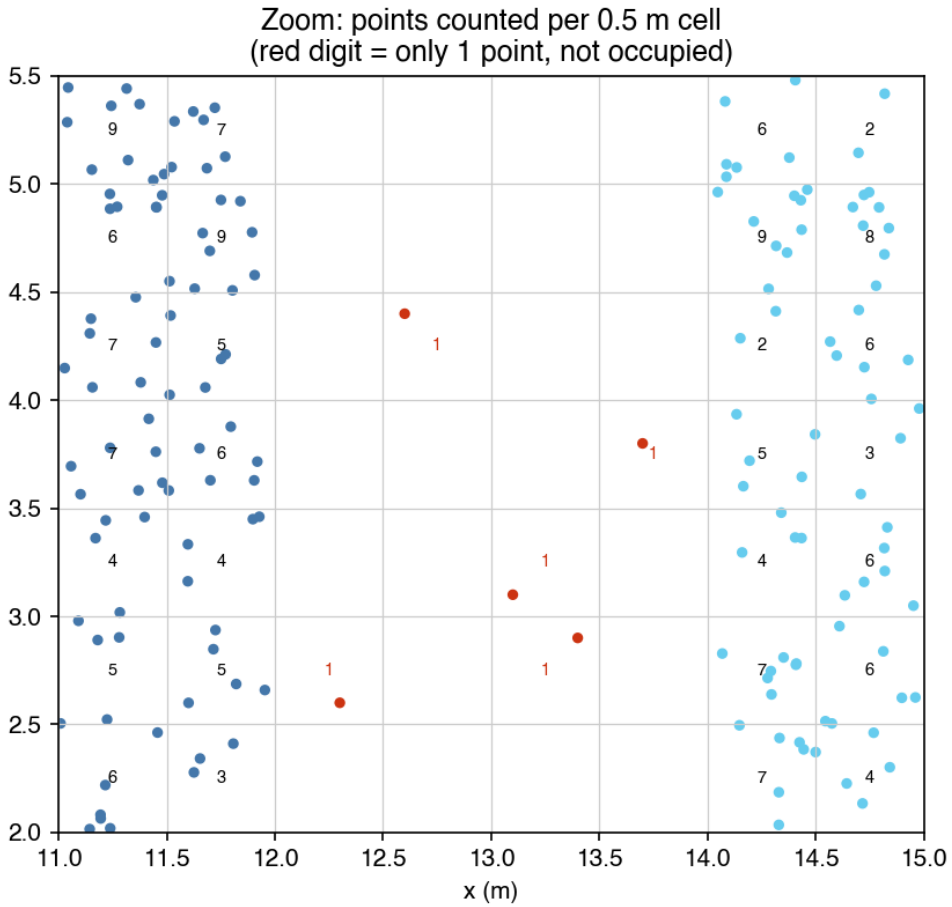
Each step is illustrated next on a 25 × 18 m synthetic scene built with the same operations and parameters.

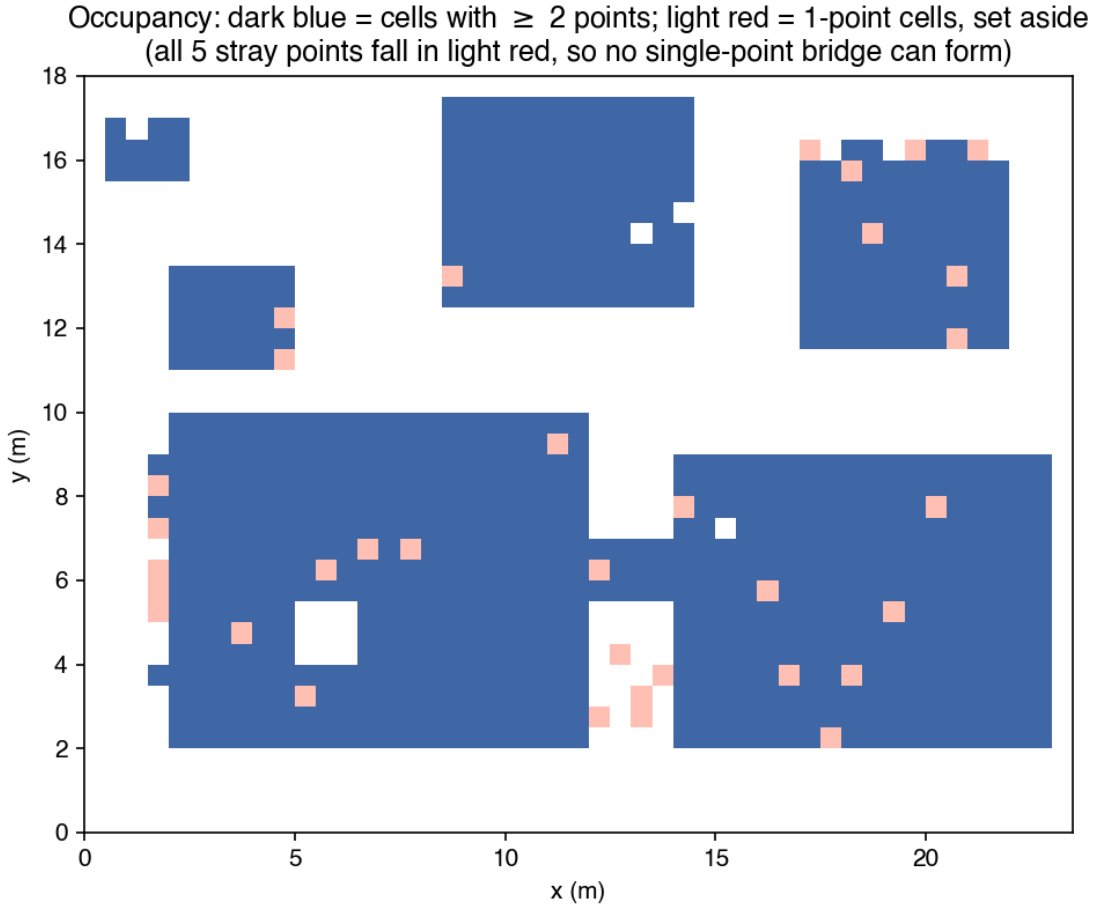
The toy scene



Two houses joined by a mislabeled tree bridge, a porch fragment, a shed, a ground-level parking lot, stray points.

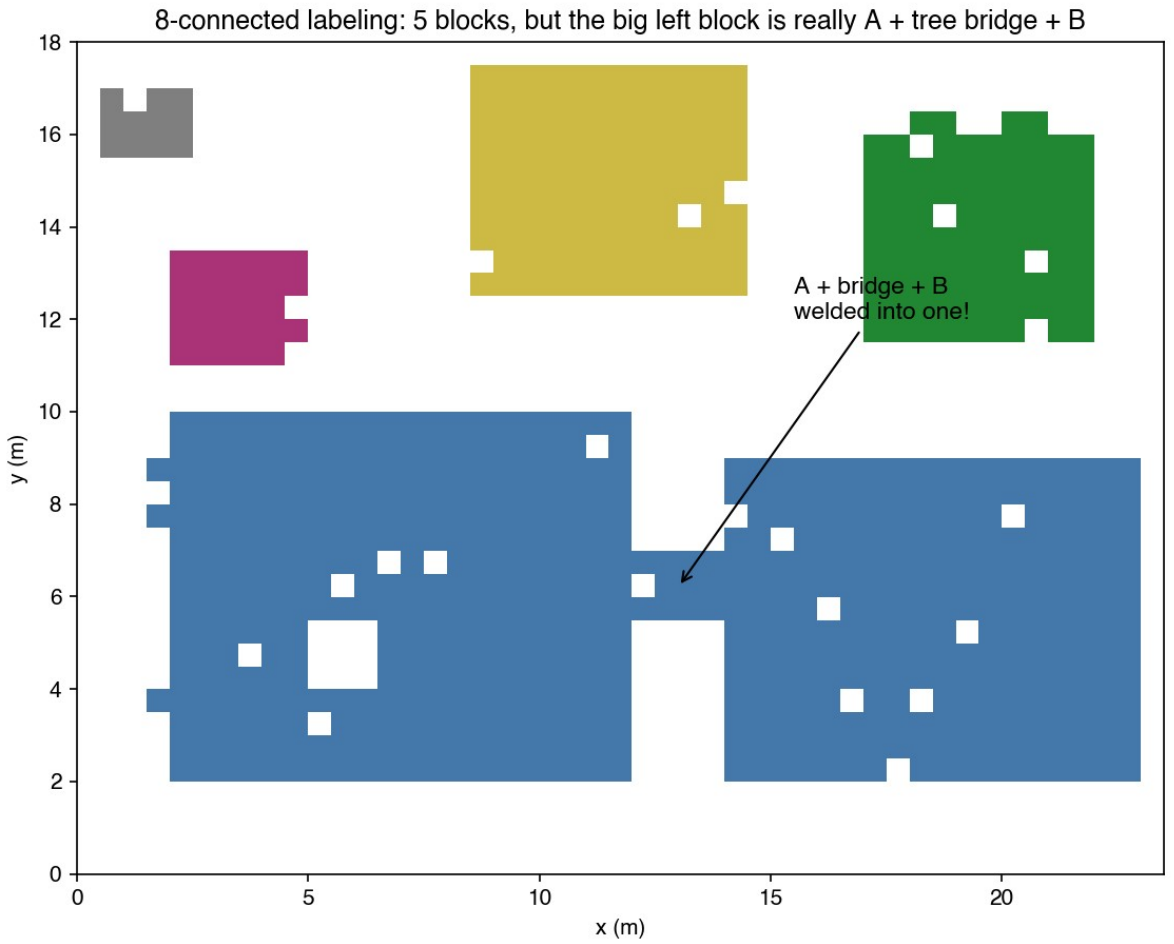
Step 2 · Rasterize: 0.5 m grid, a cell needs ≥ 2 points





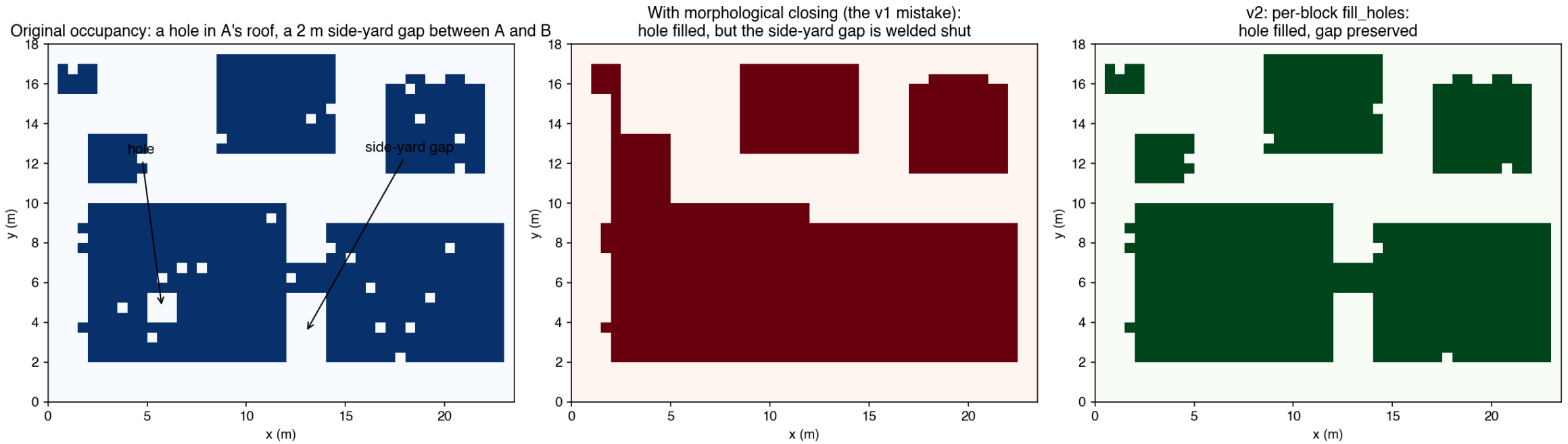
Real data: 8.6 M points collapse into one occupancy image.

Step 3 · Connected components



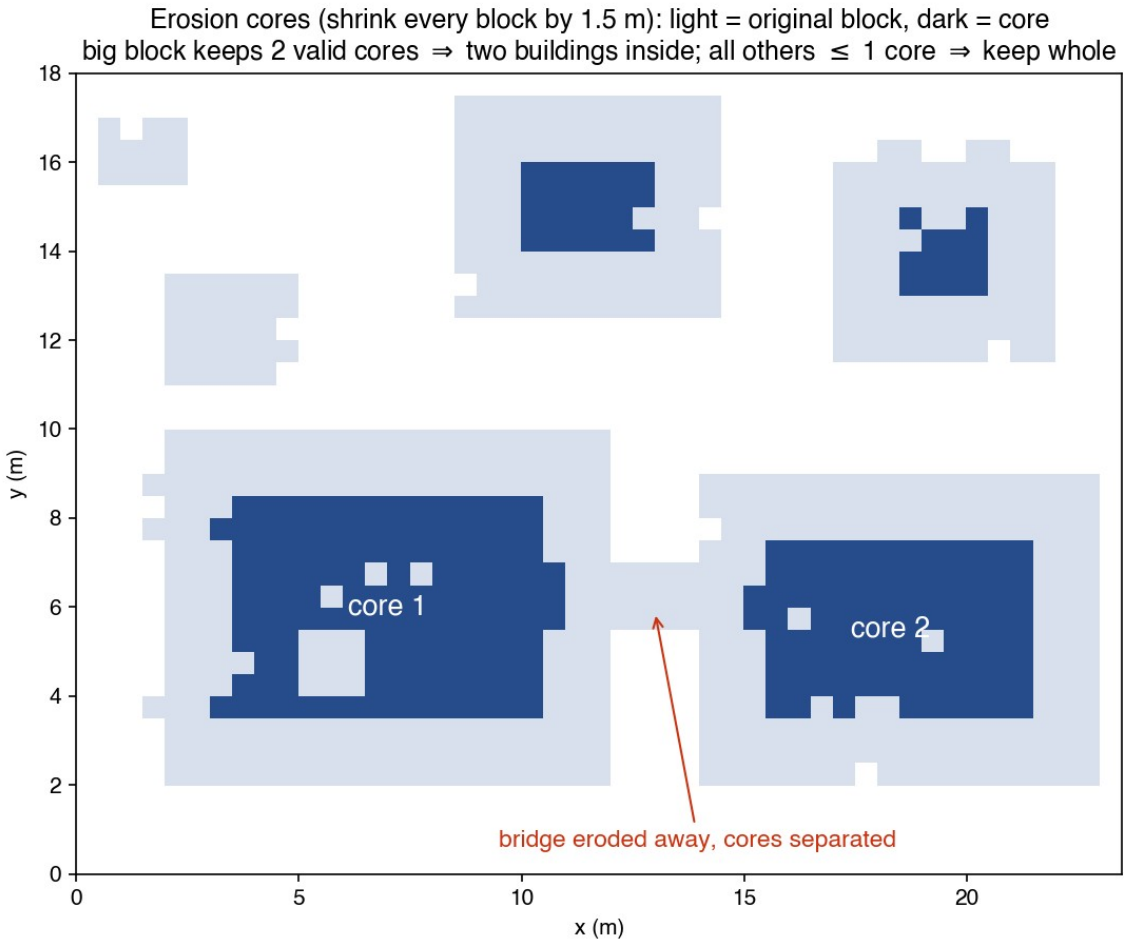
Real data: thousands of blocks — some hide several buildings behind a tree bridge.

Why not morphological closing (the v1 lesson)



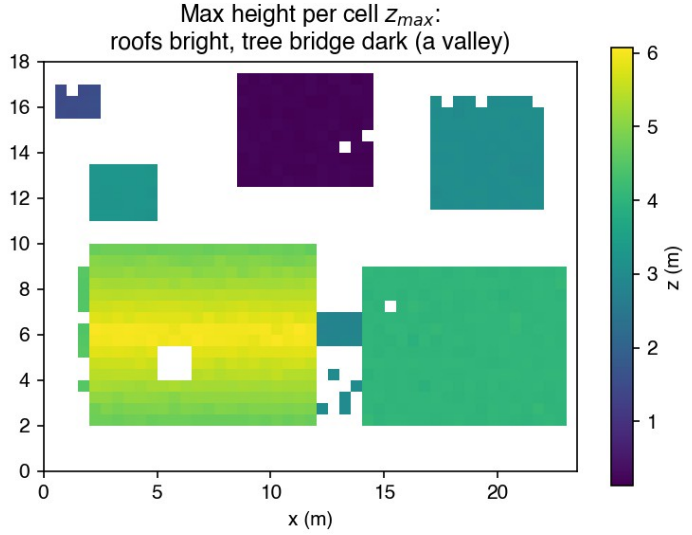
v1 used closing and welded whole rows of houses — v2 fills holes per block instead.

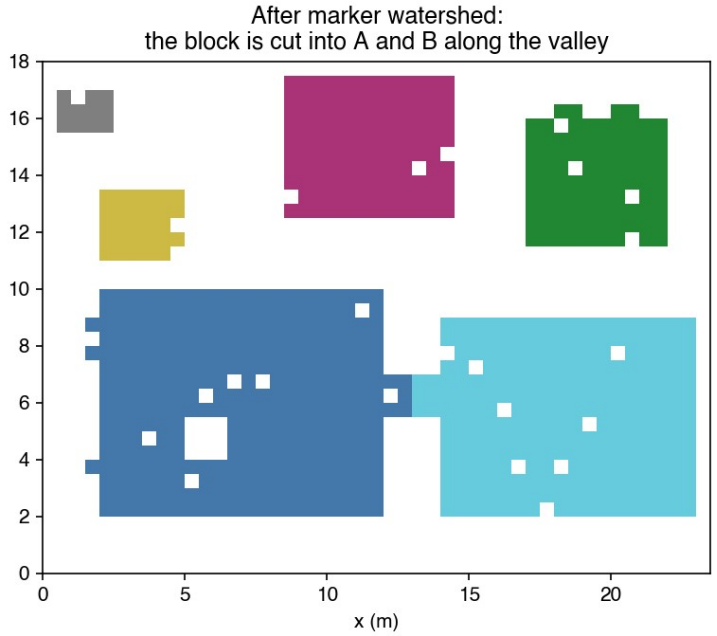
Step 4 · Erosion: how many buildings hide in a block?

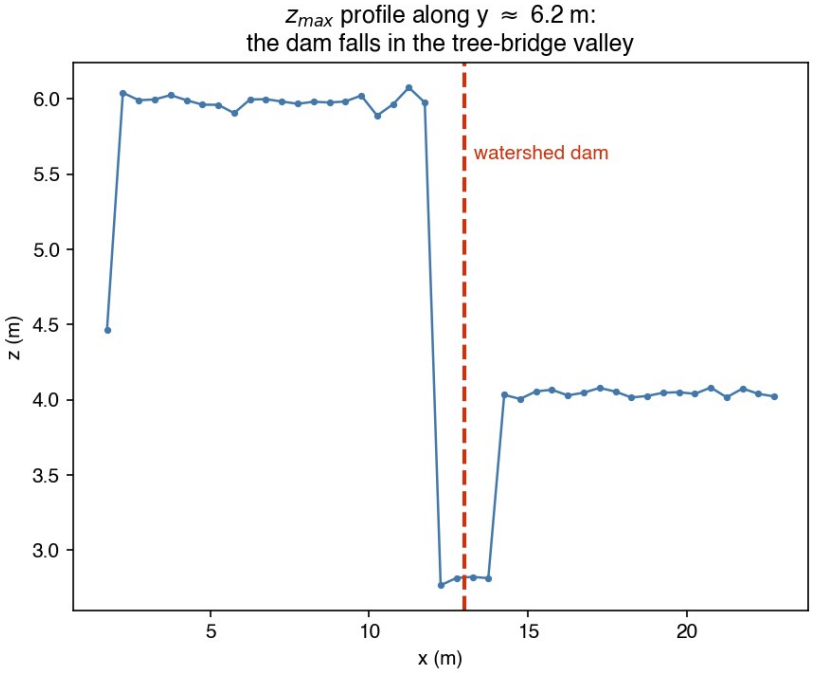


Real data: 168 blocks kept ≥ 2 cores — flagged for splitting.

Step 5 · Marker watershed: cut along the height valley

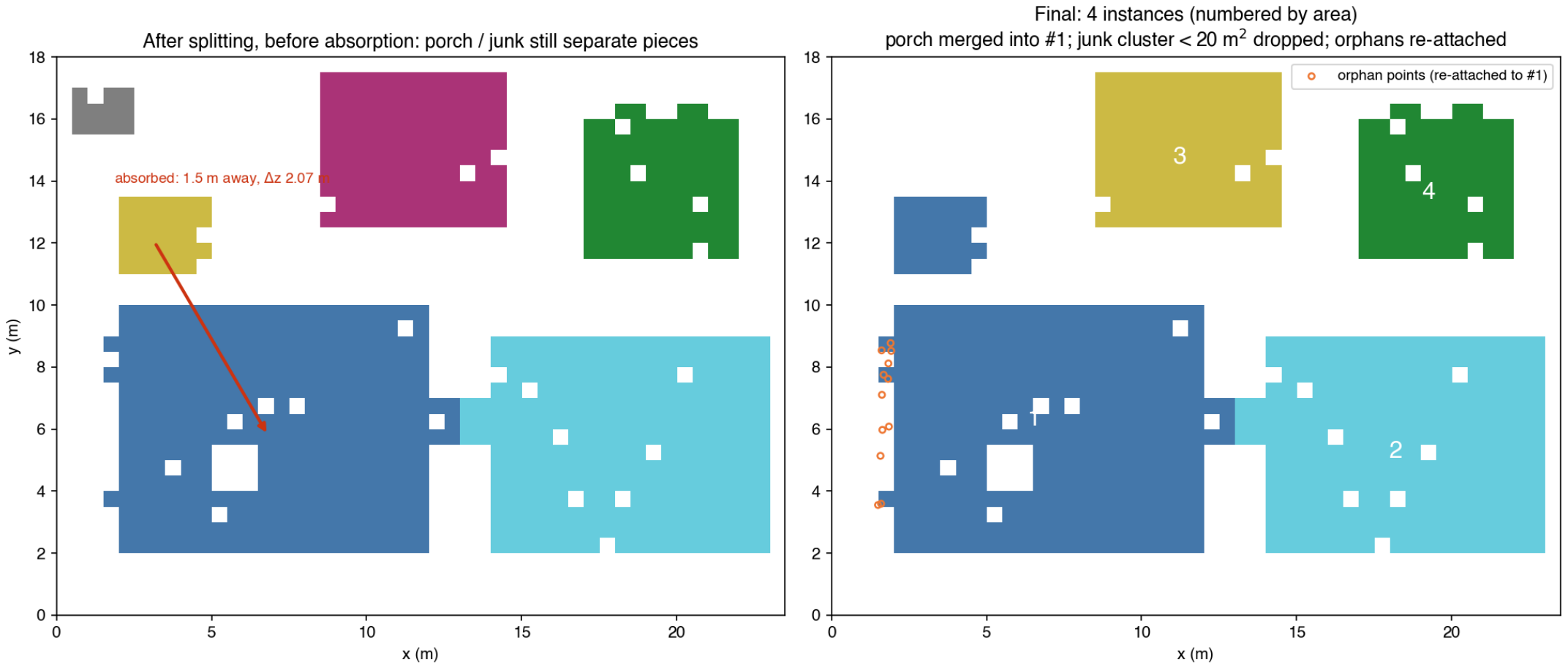






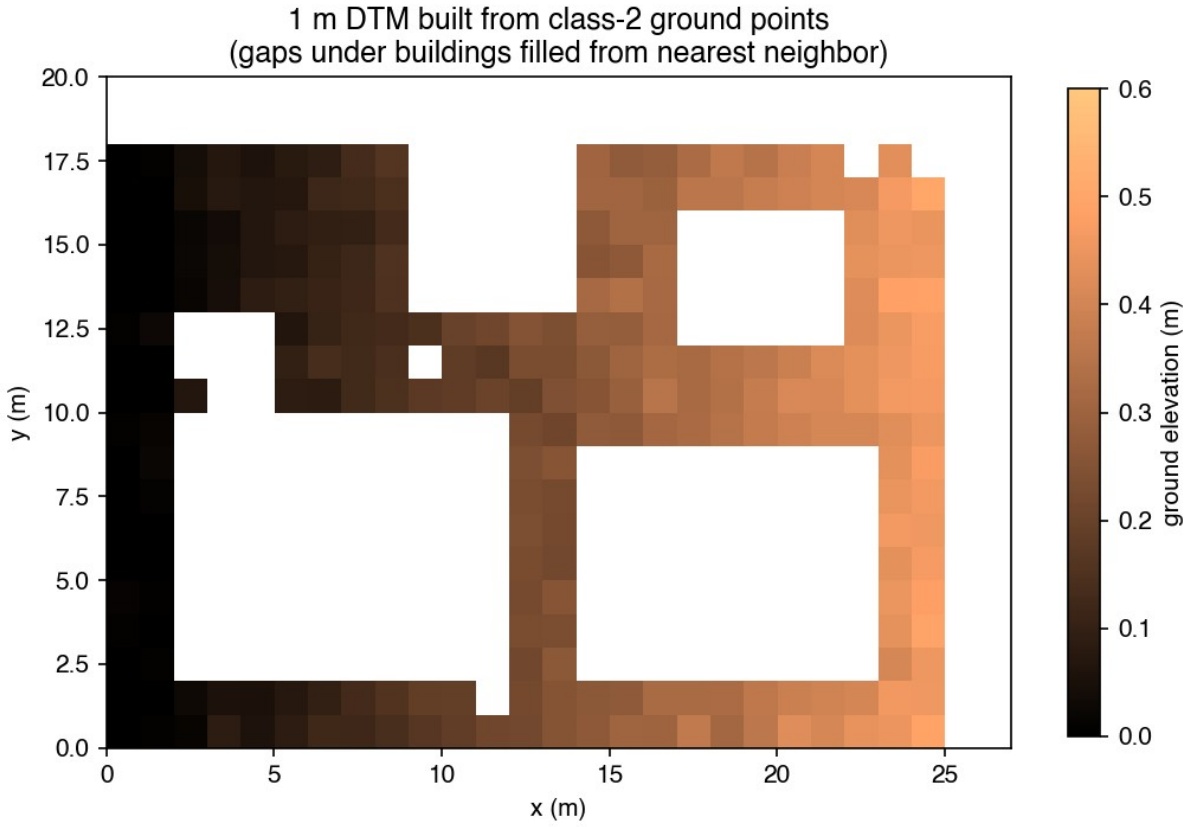
Real data: 168 merged blocks split into 383 buildings.

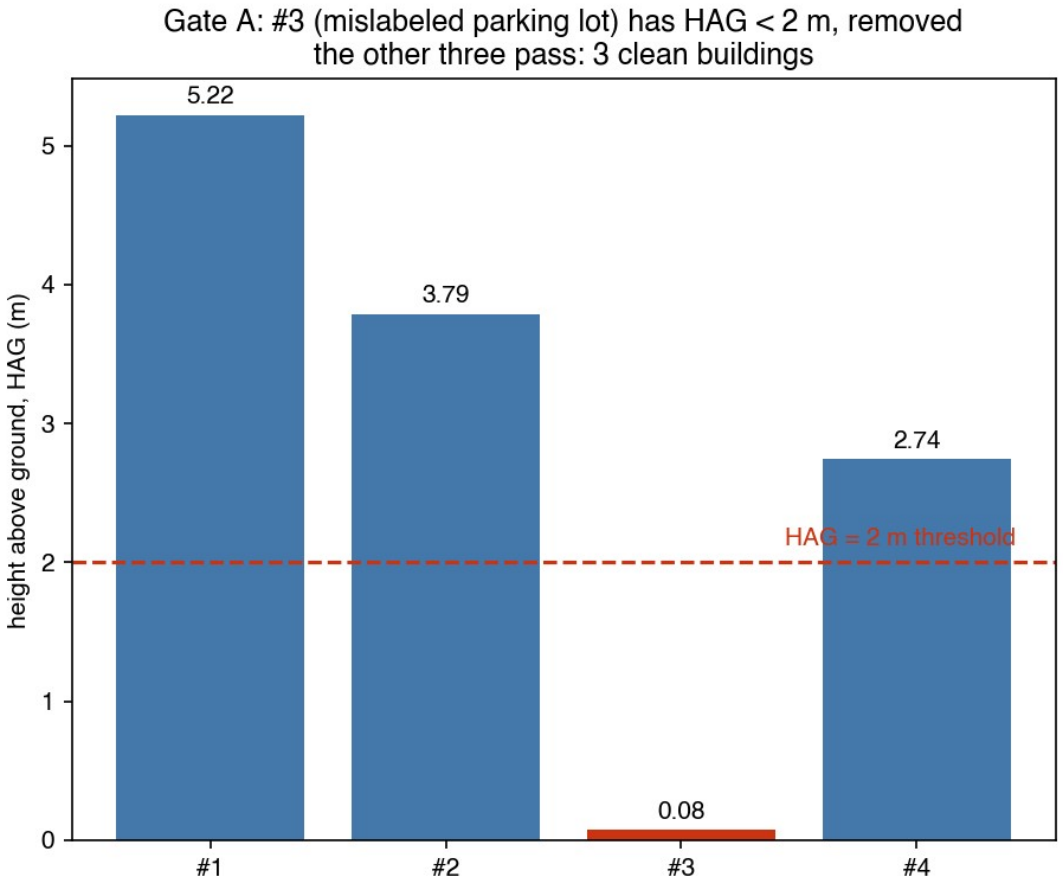
Step 6 · Fragment absorption, then final IDs



Real data: 2,296 instances after this step.

Step 7 · Gate A: height above ground kills fake buildings





Real data: shape rules + manual review removed 13 (bridges, ramps, tree rows, power lines) → 2,283 buildings.

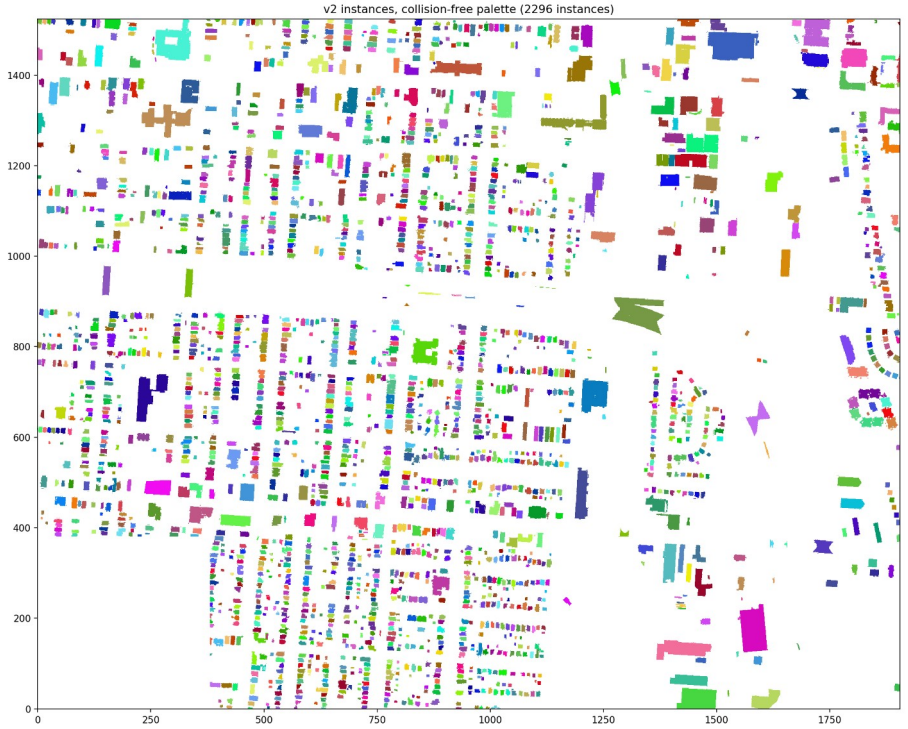
Result: 2,283 clean building instances

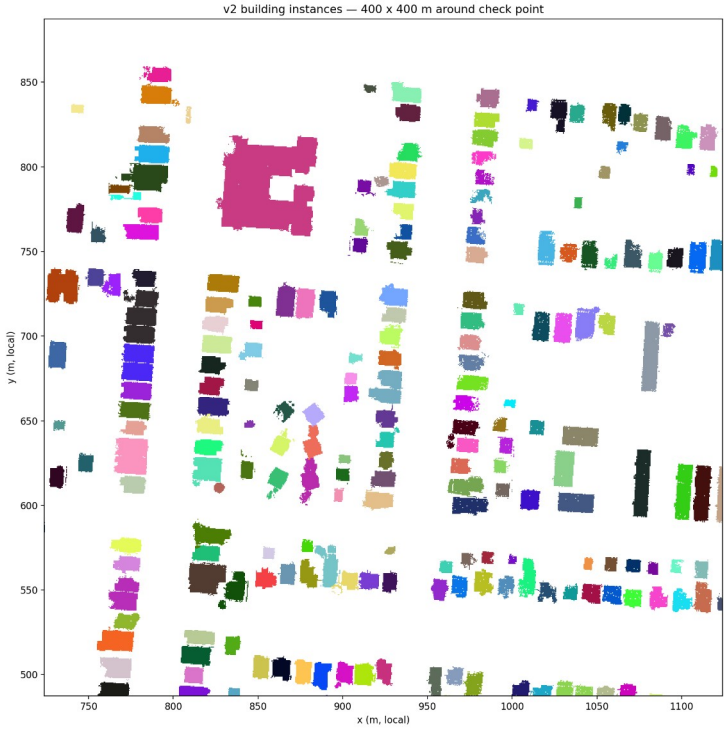
2,296
raw instances

168 → 383
merged blocks split

13
non-buildings removed

2,283
clean building instances





Same 400 × 400 m window as slide 2 — the tree-bridged houses are now separate. Collision-free palette: neighbors never share a color (verified in CloudCompare).

Next: from instances to LOD2 building models

